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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC3322-1 – REAL TIME OPERATING SYSTEMS

(For ECE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. If a real-time task misses its deadline, what should be done?
A) Always drop the task. B) Always complete the task.
C) It depends on the importance of the task. D) Restart the system.
2. What is a "joblist" in a real-time system?
A) A dynamic list of incoming user commands. B) A fixed set of application tasks or jobs preloaded into the computer.
C) A list of failed processes to be restarted. D) A record of past system executions.
3. What does "performability" of a real-time computer represent?
A) The speed of executing control algorithms. B) The probability that the system meets each accomplishment level.
C) The reliability of hardware components. D) The total number of control tasks completed.
4. What is a **straight-line code**?
A) A piece of code with multiple entry and exit points. B) A code segment executed in parallel threads.
C) A code segment with exactly one entry point and one exit point executed sequentially. D) A recursive function with a single return statement.
5. Why is the **Loop Bounds** module important in timing estimation?
A) It determines the complexity of each function call. B) It provides limits on loop iterations, which are crucial for timing prediction.
C) It maps loops to their corresponding procedures. D) It detects syntax errors in loop constructs.
6. The **relative deadline** of a task is defined as:
A) Absolute deadline minus release time B) Release time minus execution time
C) Period minus execution time D) Absolute deadline plus execution time
7. **Dynamic priority algorithms, such as Earliest Deadline First (EDF),:**
A) Assign fixed priorities to tasks B) Change task priorities over time based on deadlines
C) Do not consider deadlines D) Only work for non-preemptive schedules
8. The **Bin Packing Algorithm** in multiprocessor scheduling:
A) Assigns tasks to processors without considering utilization B) Ensures total processor utilization on each processor does not exceed a given threshold
C) Only works for aperiodic tasks D) Uses dynamic priorities exclusively
9. **Variable rate traffic** is typically associated with which type of sources?
A) Temperature sensors B) Voice and video sources
C) Clock signals D) Periodic data loggers
10. The **node degree** of a network determines:
A) Maximum message size B) Number of edges adjacent to each node
C) Network diameter D) Fault-tolerance

11. What is a key feature of circuit switching?
 - A) Each packet is independent and may follow different paths
 - B) The path is reserved exclusively for the session
 - C) Messages are broken into flits for pipelining
 - D) No connection setup is required
12. In contention-based protocols, a collision occurs when:
 - A) Nodes transmit on different channels simultaneously
 - B) Multiple nodes transmit on the same channel at the same time
 - C) Only one node transmits while others are idle
 - D) Nodes wait for the channel to become idle
13. VT-CSMA is particularly efficient under which condition?
 - A) High network load and long end-to-end delays
 - B) End-to-end transmission delay is less than the average time to transmit a packet and load is not very high
 - C) When there is only one node in the network
 - D) When nodes are not synchronized
14. Which of the following is considered an overhead in token-based algorithms?
 - A) Packet formatting time
 - B) Token capture delay
 - C) IP address assignment
 - D) Queueing delay
15. What is the primary purpose of an RTOS?
 - A) To maximize CPU throughput without considering timing constraints
 - B) To manage multitasking applications needing strict time deadlines and real-time constraints
 - C) To provide only memory management functions for embedded systems
 - D) To replace hardware-level device drivers completely
16. Which of the following describes the dynamic programming method for fixed-time scheduling?
 - A) Schedules are fixed by experience and reasoning
 - B) Timer interrupts are loaded with schedules determined by a running program
 - C) Schedules are randomly assigned
 - D) Tasks execute in round-robin order without preemption
17. In μ COS-II, which function is used to enter a critical section by disabling interrupts?
 - A) OSIntExit()
 - B) OSSchedLock()
 - C) OS_ENTER_CRITICAL
 - D) OSStart()
18. In VxWorks, which mechanism prevents a task from being deleted while holding a resource?
 - A) Mutex semaphore
 - B) Counting semaphore
 - C) Event flag
 - D) Mailbox
19. In VxWorks, which IPC method allows a task to wait for any or all conditions to be met?
 - A) Mutex semaphore
 - B) Event flags
 - C) Pipes
 - D) Queues
20. What happens if the inserted coins exceed the chocolate price in the ACVM?
 - A) The machine shuts down
 - B) The excess amount is refunded
 - C) The chocolate is not dispensed
 - D) The machine displays an error message

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Briefly explain the following RTS classes;
 - i. Periodic tasks, ii. Aperiodic tasks,
 - ii. Critical tasks, iv. Non-critical tasks
- b) Given M=4 classes. The bounds are (0.41, 1], (0.26, 0.41], (0.19, 0.26] and (0.00, 0.19] for the four classes. For the following task set given, allot the processors for each task. Compute the U assignment vector for that.

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11
e_i	5	7	3	1	10	16	1	3	9	17	21
P_i	10	21	22	24	30	40	50	55	70	90	95

8 L3 2 1

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|----|----|---|---|----|---|---|
| 2. | a) | Prove that RM algorithm is an optimal static-priority scheduling algorithm. | 8 | L2 | 2 | 1 |
| | b) | What is blocking? Explain the different types of blocking in a RTS. | 8 | L2 | 2 | 1 |
| 3. | a) | With a suitable example, explain priority inversion? Discuss the different techniques used to control it. | 8 | L2 | 2 | 1 |
| | b) | What is performability? Explain the hierarchical view of the same. | 8 | L2 | 1 | 1 |

Unit – II

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|----|----|---|---|----|---|---|
| 4. | a) | Explain the different communication media used in RTS. | 8 | L2 | 3 | 1 |
| | b) | Describe the following scheduling strategies: (i) Preemptive Scheduling. (ii) Cyclic Co-operative Round Robin Scheduling. | 8 | L2 | 4 | 1 |
| 5. | a) | Prove that in the absence of failures, the maximum cycle time of the token in timed token protocol is no greater than twice the TTRT. | 8 | L2 | 3 | 1 |
| | b) | Explain the different network architectural issues. | 8 | L2 | 3 | 1 |
| 6. | a) | List the notations used in VTCSMA. With an example explain the relation between Virtual Clock and Real Clock. | 8 | L2 | 3 | 1 |
| | b) | List and explain the OS Security functions and their actions. | 8 | L2 | 4 | 1 |

Unit – III

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|----|----|---|---|----|---|---|
| 7. | a) | Explain the RTOS System level functions used in MUCOS. | 8 | L2 | 5 | 1 |
| | b) | What are the features of VxWorks that makes a good selection as an RTOS? | 8 | L2 | 5 | 1 |
| 8. | a) | Draw a neat state diagram of the TCP/IP stack and explain the same. | 8 | L2 | 5 | 1 |
| | b) | With a neat sketch, explain the requirements of Automatic Chocolate Vending System. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
